

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/01

Paper 1 Pure Mathematics 1 **(P1)**

May/June 2005

1 hour 45 minutes

Additional materials: Answer Booklet/Paper
Graph paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
At the end of the examination, fasten all your work securely together.

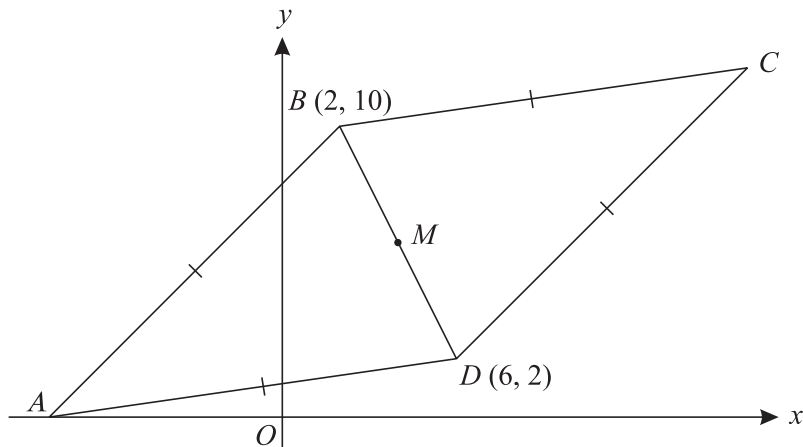
The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 75.
Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

This document consists of **4** printed pages.



- 1 A curve is such that $\frac{dy}{dx} = 2x^2 - 5$. Given that the point (3, 8) lies on the curve, find the equation of the curve. [4]
- 2 Find the gradient of the curve $y = \frac{12}{x^2 - 4x}$ at the point where $x = 3$. [4]
- 3 (i) Show that the equation $\sin \theta + \cos \theta = 2(\sin \theta - \cos \theta)$ can be expressed as $\tan \theta = 3$. [2]
 (ii) Hence solve the equation $\sin \theta + \cos \theta = 2(\sin \theta - \cos \theta)$, for $0^\circ \leq \theta \leq 360^\circ$. [2]
- 4 (i) Find the first 3 terms in the expansion of $(2 - x)^6$ in ascending powers of x . [3]
 (ii) Find the value of k for which there is no term in x^2 in the expansion of $(1 + kx)(2 - x)^6$. [2]

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The diagram shows a rhombus $ABCD$. The points B and D have coordinates (2, 10) and (6, 2) respectively, and A lies on the x -axis. The mid-point of BD is M . Find, by calculation, the coordinates of each of M , A and C . [6]

- 6 A geometric progression has 6 terms. The first term is 192 and the common ratio is 1.5. An arithmetic progression has 21 terms and common difference 1.5. Given that the sum of all the terms in the geometric progression is equal to the sum of all the terms in the arithmetic progression, find the first term and the last term of the arithmetic progression. [6]

7 A function f is defined by $f : x \mapsto 3 - 2 \sin x$, for $0^\circ \leq x \leq 360^\circ$.

(i) Find the range of f . [2]

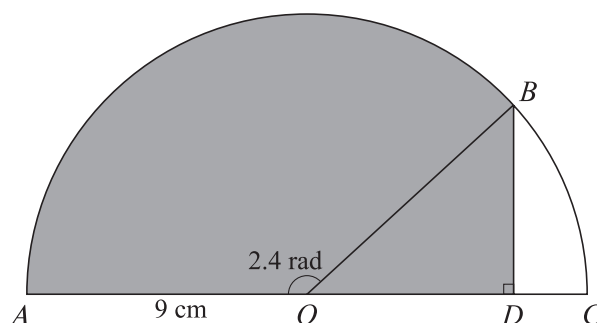
(ii) Sketch the graph of $y = f(x)$. [2]

A function g is defined by $g : x \mapsto 3 - 2 \sin x$, for $0^\circ \leq x \leq A^\circ$, where A is a constant.

(iii) State the largest value of A for which g has an inverse. [1]

(iv) When A has this value, obtain an expression, in terms of x , for $g^{-1}(x)$. [2]

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In the diagram, ABC is a semicircle, centre O and radius 9 cm. The line BD is perpendicular to the diameter AC and angle $AOB = 2.4$ radians.

(i) Show that $BD = 6.08$ cm, correct to 3 significant figures. [2]

(ii) Find the perimeter of the shaded region. [3]

(iii) Find the area of the shaded region. [3]

9 A curve has equation $y = \frac{4}{\sqrt{x}}$.

(i) The normal to the curve at the point $(4, 2)$ meets the x -axis at P and the y -axis at Q . Find the length of PQ , correct to 3 significant figures. [6]

(ii) Find the area of the region enclosed by the curve, the x -axis and the lines $x = 1$ and $x = 4$. [4]

10 The equation of a curve is $y = x^2 - 3x + 4$.

(i) Show that the whole of the curve lies above the x -axis. [3]

(ii) Find the set of values of x for which $x^2 - 3x + 4$ is a decreasing function of x . [1]

The equation of a line is $y + 2x = k$, where k is a constant.

(iii) In the case where $k = 6$, find the coordinates of the points of intersection of the line and the curve. [3]

(iv) Find the value of k for which the line is a tangent to the curve. [3]

11 Relative to an origin O , the position vectors of the points A and B are given by

$$\overrightarrow{OA} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k} \quad \text{and} \quad \overrightarrow{OB} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}.$$

(i) Use a scalar product to find angle AOB , correct to the nearest degree. [4]

(ii) Find the unit vector in the direction of $\overrightarrow{AB}$. [3]

(iii) The point C is such that $\overrightarrow{OC} = 6\mathbf{j} + p\mathbf{k}$, where p is a constant. Given that the lengths of $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are equal, find the possible values of p . [4]